International Journal of Trend in Research and Development, Volume 4(4), ISSN: 2394-9333
www.ijtrd.com

Use of Recycled Concrete Aggregate and its Influence on Strengths of Concrete

[1]Nikhil V. Mulik, [2]Sudhir L. Bhilare and [3]G. R. Patil,
[1]P.G. Student, [2,3]Associate Professor,
[1,2,3]Department of Civil Engineering, J.S.P.M's R.S.C.O.E, Tathawade, Pune, Maharashtra, India

***Abstract*—**In our present society the major challenges is protection of environment. Cities are suffering because of their own Construction and Demolition Waste (CDW) with serious environmental consequences. Reducing, recycling and re-using appear to be the best option. Concrete deterioration may be due to many reasons: Corrosion of enclosed metals, Chemical attack, Alkali-Aggregate reactivity, Fire, Overloaded, etc.

The techniques for producing concrete using recycled aggregate obtained from CDW. Also the effects are studied on the compressive strength and flexure strength of concrete conducting various laboratory trials, replacing the natural aggregate by the recycled ones. Replacement of aggregates is done in percentage.

So in this project, concrete was tested made from recycled aggregates replacing natural aggregates. Such that the recycled aggregates were sub-graded, with respect to the service life of the parent concrete.

Finally, the investigation and comparison of experimental results of strengths of natural aggregate concrete and recycled aggregate concrete with the help of statistical data obtained by testing of specimen was done, as per recommended procedures by IS 516:1959.

Keywords*—Construction and Demolition Waste (CDW), Natural Aggregate (NA), Recycled Aggregates (RA), Compressive Strength, Flexure Strength, Concrete.*

I. INTRODUCTION

In India Construction has accounted for around 40 per cent of the development investment during the past 50 years. Around 16 per cent of the nation's working population depends on construction for its livelihood. It contributes more than 5 per cent to the nation's gross domestic product (GDP) and 78 per cent to the gross capital formation (GCF). The share of the Indian construction sector in total GCF came down from 60 per cent in 1970-71 to 34 per cent in 1990-91. Thereafter, it increased to 48 per cent in 1993-94 and stood at 44 per cent in 1999-2000. In 21st century, there has been an increase in the share of the construction sector in GDP and GCF.

Concrete, one of the most widely used construction material, has been subjected to major research and development over the past century. Once thought to provide sound structural and durable performance with little else, concrete has now evolved into a material that is capable of being suitable for environment.

Through the development of high strength concrete during the early sixties for rapid urbanization and high strength requirement, the performance based on concrete mix design was established. High strength concretes are now not only used for their strength capabilities, but also for the enhanced durability characteristics.

Any construction activity requires several materials such as concrete, steel, brick, stone, glass, clay, mud, wood, and so on. However, the cement concrete remains the main construction material used in construction industries. For its suitability and adaptability with respect to the changing environment, the concrete must be such that it can conserve resources, protect the environment, economize and lead to proper utilization of energy. To achieve this, major emphasis must be laid on the use of wastes and byproducts in cement and concrete used for new constructions.

The use of recycled aggregates from construction and demolition wastes is showing prospective application in construction as alternative to natural aggregates. It conserves natural resources and reduces the space required for the landfill disposal.

According to Central Public Health & Environmental Engineering Organization (CPHEEO), India produces 10-12 million tons of CDW per annum. Amongst which Soil, Sand & Gravels is 36%, Brick & Masonry is 31%, 23% of Concrete, 5% Metals, 2% each of Bitumen and Wood and 1% for others. India is even facing a shortage of about 55,000 million cubic meters of aggregates for just the real estate industry. Also in road construction sector about 750 million cubic meters of aggregate is required to achieve the target. If there are no favourable uses, the materials become wastes and there will be a large environmental and financial pressure on the authorities to treat and dispose them.

II. TEST SPECIMEN

A. *Material used*

- *Cement:* Ordinary Portland cement (OPC) of 43 grade of PriyaCement is used in this experimental work. Weight of each cement bag is 50 kg. The standard consistency of cement sample is found to be 35.3%. The setting time for the given cement sample are found to be:

  Initial setting time = 42 min
  Final setting time = 520 min

- *Fine Aggregates:* Crushed sand having Specific Gravity 2.32gm/cc and Fineness Modulus as 4.212.
- *Natural Coarse Aggregate:* Consist of 20 mm crushed aggregate,having Specific Gravity 2.59gm/cc and Fineness Modulus as 6.45.
- *Recycled Coarse Aggregate:* Consist of 20mm crushed concrete. Recycled Coarse Aggregate was collected from different site according to the life spans of the building.

  B1– 7 Years
  Near Flying Officer SudhirPawar Ln,
  NarveerTanajiWadi, Shivajinagar,
  Pune, Maharashtra 411005.

  B2– 21 Years
  Sector 18A, New Panvel East, Panvel,
  Navi Mumbai, Maharashtra 410206.

B3 – 61 Years
Near Fergusson College Campus,
Shivajinagar, Pune, Maharashtra 411004

B4 – 12 Years
Near Petrol Pump, Hedgewar Path,
Adarsh Nagar, Kalewadi,
Pimpri-Chinchwad, Maharashtra 411017

B. Mix proportion and casting procedure

Throughout hand mixing over a tray was done. Coarse aggregate with the percent wise replacement of recycled aggregate were put in the tray first, after that fine aggregate followed by cement were added. The materials were first mixed thoroughly for 1 minute in dry condition itself. Water was added while the mixing was in continuation. Concrete was then placed in IS specified moulds in three layers, each layer was being compacted by standard tamping rod with more than 35 strokes. Exposed surface was finished with trowel to avoid uneven surface. A total of 246 (i.e. 60 cubes of B1, 60 cubes of B2, 60 cubes of B3, 60 cubes of B4 and 6 cubes of conventional concrete) concrete specimens of dimension 150x150x150mm for compressive strength testand similarly,246 concrete specimens of dimension 100x100x500mm flexural strength testwere designed and fabricated. Specimens were prepared to obtain characteristic cube strength of 30 MPa. The natural aggregates were replaced by percentage variation i.e. 0, 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100%.In particular, 3 specimens for compressive strength test and 3 specimens for flexural strength test, for each percentage variation. Specimens were cured by immersing them in curing tank and tested for7th and28thdays.

TABLE I. MIX PROPORTION AND CONCRETE BATCHES(FOR CONVENSIONAL CONCRETE)

Grade	Cement (kg/m^3)	Aggregate (kg/m^3)		Water (lit/m^3)
		Fine	*Coarse*	
M30	466	346.11	1362.01	177

TABLE II. MIX PROPORTION AND CONCRETE BATCHES (FOR B1 AND B2)

Percentage Variation	Cement (kg/m^3)	Aggregate (kg/m^3)			Water (lit/m^3)
		Fine	*Coarse*	*Recycled*[a]	
10	466	346.11	1225.81	136.20	177
20	466	346.11	1089.61	272.40	177
30	466	346.11	953.41	408.60	177
40	466	346.11	817.21	544.80	177
50	466	346.11	681.00	681.00	177
60	466	346.11	544.80	817.21	177
70	466	346.11	408.60	953.41	177
80	466	346.11	272.40	1089.61	177
90	466	346.11	136.20	1225.81	177
100	466	346.11	0	1362.01	177

a. Only coarse aggregate proportion is changed as per percentage, else is kept the same

III. TEST PROGRAM

A. Compression testing of cubes

Cubes as casted of size 150 x 150 x 150mm were tested using Compression testing machine (CTM) of capacity 250 ton, capable of giving load at the rate of 140 kg/sq.cm/min. Testing of the cubes wasdone at the age of 7th and 28thday. The cubes were placed in the machine between wiped and cleaned loading surfaces and load is given approximately at the rate of 140 kg/sq.cm/min. and ultimate crushing load is noted to calculate crushing strength of concrete according to IS: 516-1959.

The measuring strength of specimen is calculated by dividing the maximum load applied to the specimen during the test by the cross section area.

B. Universal Testing Machine

Beams as casted of size 100 x 100 x 500 mm were tested using Universal Testing Machine (UTM) of capacity 60 ton. Testing of beams was done at the age of 7th and 28th day. Specimen was supported on two steel rollers, which were so placed that the distance from centre to centre is 400 mm. The loading force is applied in the middle by a steel roller. All rollers shall be mounted in such a manner that the load is applied axially and without subjecting the specimen to any torsional stresses or restraints.

The measuring strength of specimen is calculated by the following formula,

$$\frac{3LF}{2bd^2} \quad (1)$$

Where,

L - Length of specimen
F - Total force applied to the specimen
b- Width of specimen
d - Thickness of specimen

IV. RESULTS AND DISCUSSION

Following are the obtained results of Compression Test and Flexural Test.

A. Compressive Strength

TABLE III. COMPRESSIVESTRENGTH FOR CONVENTIONAL CONCRETE

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
0	21.76 N/mm^2	32.43 N/mm^2

Average strength of the three cubes

TABLE IV. COMPRESSIVE STRENGTH FOR B1

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	18.64 N/mm^2	28.24 N/mm^2
20	19.51 N/mm^2	24.29 N/mm^2
30	18.12 N/mm^2	24.68 N/mm^2
40	17.9 N/mm^2	25.06 N/mm^2
50	16.02 N/mm^2	23.19 N/mm^2
60	13.45 N/mm^2	19.72 N/mm^2
70	11.93 N/mm^2	19.21 N/mm^2
80	11.54 N/mm^2	17.95 N/mm^2
90	9.63 N/mm^2	15.26 N/mm^2
100	7.21N/mm^2	13.47 N/mm^2

Average strength of the three cubes

TABLE V. COMPRESSIVE STRENGTH FOR B2

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	20.45 N/mm^2	29.58 N/mm^2
20	21.14 N/mm^2	31.55 N/mm^2
30	19.87 N/mm^2	30.48 N/mm^2
40	19.36 N/mm^2	28.41 N/mm^2
50	18.97 N/mm^2	26.19 N/mm^2
60	17.23 N/mm^2	25.93 N/mm^2
70	16.04 N/mm^2	21.72 N/mm^2
80	13.27 N/mm^2	16.46 N/mm^2
90	12.18 N/mm^2	16.22 N/mm^2
100	10.01 N/mm^2	14.27 N/mm^2

Average strength of the three cubes

TABLE VI. COMPRESSIVE STRENGTH FOR B3

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	16.68 N/mm^2	24.55 N/mm^2
20	15.88 N/mm^2	19.82 N/mm^2
30	10.74 N/mm^2	17.64 N/mm^2
40	7.21 N/mm^2	15.93 N/mm^2
50	6.99 N/mm^2	11.72 N/mm^2
60	6.41 N/mm^2	10.09 N/mm^2
70	5.72 N/mm^2	7.99 N/mm^2
80	5.14 N/mm^2	7.57 N/mm^2
90	4.87 N/mm^2	7.17 N/mm^2
100	4.16 N/mm^2	6.12 N/mm^2

Average strength of the three cubes

TABLE VII. COMPRESSIVE STRENGTH FOR B4

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	22.04 N/mm^2	30.55 N/mm^2
20	20.45 N/mm^2	29.61 N/mm^2
30	19.86 N/mm^2	28.57 N/mm^2
40	19.58 N/mm^2	31.69 N/mm^2
50	20.17 N/mm^2	29.16 N/mm^2
60	18.67 N/mm^2	28.31 N/mm^2
70	17.54 N/mm^2	27.73 N/mm^2
80	16.8 N/mm^2	25.18 N/mm^2
90	15.78 N/mm^2	24.92 N/mm^2
100	16.08 N/mm^2	23.51 N/mm^2

Average strength of the three cubes

Graph will be plotted with percentage variation on abscissa and Compressive Strength of concrete on ordinate.

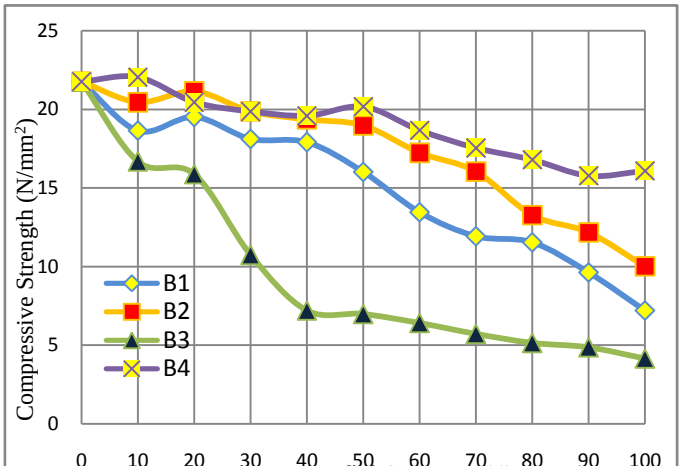


Fig. 1. Comparison of Compressive Strengths at 7th day

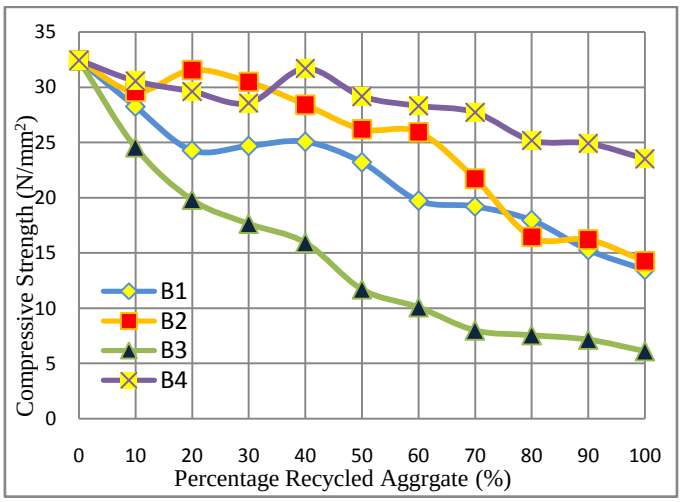


Fig. 2. Comparison of Compressive Strengths at 28th day

From the results obtained, it was observed that the results of compressive strength of recycled aggregate concrete is satisfactory during early stage of replacement but it exhibits lower compressive strength during later stage of replacement.

The Figure 1 compares the compressive strengths on 7th day of the concretes made of B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. It was seen that the concrete gives a satisfactory value of compressive strength till 20% replacement of natural aggregate by recycled aggregate.

Similarly, Figure 2 compares the compressive strengths on 28th day of the concretes made B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. Here it came across that results of compressive strength of concrete upto40% replaced aggregate were satisfactory.

It was found that aggregate obtained from B1 site, have not just solid aggregate but a large proportion of harden mortar. As harden mortar cannot bear load, the complete concrete collapses at lower compressive strength. But as we use older concrete, the aggregate obtained from it have lesser proportion of harden mortar. Deterioration of concrete is more as the time passes. This gave much better results than B1 i.e. young RA.

This can be corrected by using proper cleaning techniques and casting methods under highly skilled supervision.

B. Flexural Strength

TABLE VIII. FLEXURALSTRENGTH FOR CONVENTIONAL CONCRETE

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
0	2.90 N/mm^2	3.57 N/mm^2

Average strength of the three Beams

TABLE IX. FLEXURAL STRENGTH FOR B1

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	2.68 N/mm^2	3.33 N/mm^2
20	2.75 N/mm^2	3.09 N/mm^2
30	2.68 N/mm^2	3.11 N/mm^2
40	2.63 N/mm^2	3.13 N/mm^2
50	2.41 N/mm^2	3.02 N/mm^2
60	2.28 N/mm^2	2.78 N/mm^2
70	2.14 N/mm^2	2.74 N/mm^2
80	2.11 N/mm^2	2.65 N/mm^2
90	1.84 N/mm^2	2.45 N/mm^2
100	1.66 N/mm^2	2.30 N/mm^2

Average strength of the three Beams

TABLE X. FLEXURAL STRENGTH FOR B2

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	2.81 N/mm^2	3.41 N/mm^2
20	2.87 N/mm^2	3.52 N/mm^2
30	2.76 N/mm^2	3.46 N/mm^2
40	2.7 N/mm^2	3.34 N/mm^2
50	2.71 N/mm^2	3.20 N/mm^2
60	2.58 N/mm^2	3.19 N/mm^2
70	2.51 N/mm^2	2.92 N/mm^2
80	2.26 N/mm^2	2.54 N/mm^2
90	2.17 N/mm^2	2.52 N/mm^2
100	1.96 N/mm^2	2.37 N/mm^2

Average strength of the three Beams

TABLE XI. FLEXURAL STRENGTH FOR B3

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	2.56 N/mm^2	3.10 N/mm^2
20	2.49 N/mm^2	2.79 N/mm^2
30	2.05 N/mm^2	2.63 N/mm^2
40	1.68 N/mm^2	2.50 N/mm^2
50	1.66 N/mm^2	2.14 N/mm^2
60	1.59 N/mm^2	1.99 N/mm^2
70	1.50 N/mm^2	1.77 N/mm^2
80	1.42 N/mm^2	1.72 N/mm^2
90	1.38 N/mm^2	1.68 N/mm^2
100	1.28 N/mm^2	1.55 N/mm^2

Average strength of the three Beams

TABLE XII. FLEXURAL STRENGTH FOR B4

Percentage Variation (%)	Curing Age	
	7th Day	*28th Day*
10	2.92 N/mm^2	3.46 N/mm^2
20	2.81 N/mm^2	3.41 N/mm^2
30	2.77 N/mm^2	3.35 N/mm^2
40	2.75 N/mm^2	3.52 N/mm^2
50	2.79 N/mm^2	3.38 N/mm^2
60	2.69 N/mm^2	3.33 N/mm^2
70	2.60 N/mm^2	3.30 N/mm^2
80	2.55 N/mm^2	3.14 N/mm^2
90	2.47 N/mm^2	3.13 N/mm^2
100	2.49 N/mm^2	3.04 N/mm^2

Average strength of the three Beams

Graph will be plotted with percentage variation on abscissa and Flexural Strength of concrete on ordinate.

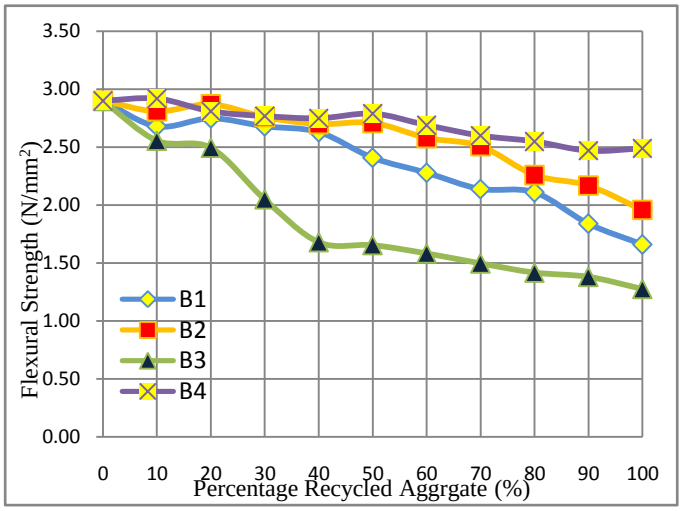


Fig. 3. Comparison of Flexural Strengths at 7th day

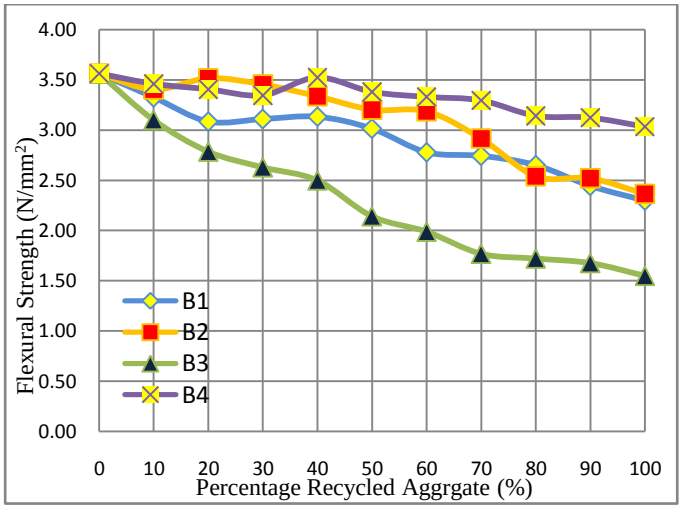


Fig. 4. Comparison of Flexural Strengths at 28th day

From the results obtained, it was observed that the results of flexural strength of recycled aggregate concrete is satisfactory during early stage of replacement but it exhibits lower flexural strength during later stage of replacement.

The Figure 3 compares the flexural strengths on 7th day of the concretes made of B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. It was seen that the concrete gives a satisfactory value of flexural strength till 20% replacement of natural aggregate by recycled aggregate.

Similarly, Figure 4 compares the flexural strengths on 28th day of the concretes made ofB1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. Here it came across that results of flexural strength of concrete upto10% replaced aggregate were satisfactory.

This can be corrected by using proper cleaning techniques and casting methods under highly skilled supervision.

CONCLUSION

From the experimental work carried out the following conclusion can be drawn:

[1] Recycled Aggregate from intermediate age of building gives better results, as deterioration of concrete is more with time and aggregates obtained are more solid.
[2] Workability decreases with the increase in percentage replacement of recycled aggregate.
[3] Tests conducted on recycled aggregates concrete and results compared with natural aggregates concrete are satisfactory.
[4] Compressive strength and flexural strength of concrete with 30% replaced recycled aggregate were acceptable.
[5] It was observed that if we requirement is of a particular strength of concrete using recycled aggregate, one should adopt a higher proportion of mix design than the required.
[6] Use of recycled aggregate, cost of transportation of natural resources & excavation is considerably saved. This reduces the waste material impact on environment.
[7] As the treatment process for recycled aggregate was inadequate, strength achieved were not upto mark, but by applying more advanced treatment process the strength can be improved.

Acknowledgment

The outcome has been achieved with the support of my guide Prof. S. L. Bhilare, my P.G. Coordinator Prof. G. RPatil, R.S.C.O.E., Tathawade. Special thanks to S.R.T.T.C, Faculty of Engineering, Khamset for granting me permission to conduct my experimental work. Finally, I thanks to my parents, and my classmates to support me mentally and physically.

References

[1] Dr. Sunil S. Pimplikar, "Use of recycled aggregate concrete", IOSR Journal of Mechanical and Civil Engineering (IOSR-JMCE) ISSN: 2278-1684, PP: 52-59.
[2] MirjanaMalešev, VlastimirRadonjanin and SnežanaMarinković, "Recycled concrete as aggregate for structural concrete production", ISSN 2071-1050.
[3] Margaret Mary O'Mahony, "Recycling of material in civil engineering", University of Oxford, 1990.
[4] David R. Wilburn and Thomas G. Goonan, "Aggregates from natural and recycled sources, economic assessments for construction applications—a materials flow analysis", U.S. GEOLOGICAL SURVEY CIRCULAR 1176, June 1, 1998.
[5] The Government of Hong Kong special administrative region, "Durability study for normal grade concrete with recycle aggregates", Version 1, Last printed 15/5/2007.
[6] Cement Concrete & Aggregates Australia, "Use of recycled aggregates in construction", May 2008.
[7] Ashraf M. Wagih, Hossam Z. El-Karmoty, Magda Ebid, Samir H. Okba, "Recycled construction and demolition concrete waste as aggregate for structural concrete", Housing and Building National Research Center HBRC Journal. Received 19 July 2012; accepted 24 September 2012.
[8] Kho Pin Verian, Nancy M. Whiting, Jan Olek, Jitendra Jain and Mark B. Snyder, "Using recycled concrete as aggregate in concrete pavements to reduce materials cost", Report No. - FHWA/IN/JTRP-2013/18, August 2013.
[9] SudhirP.Patil, Ganesh S.Ingle and PrashantD.Sathe, "Recycled coarse aggregates", International Journal of Advanced Technology in Civil Engineering, ISSN: 2231 –5721, Volume-2, Issue-1, 2013.
[10] Yasser AbdelghanyFawzy, "Impact of recycled gravel obtained from low or medium concrete grade on concrete properties", Housing and Building National Research Center HBRC Journal, Received 22 September 2015; revised 9 April 2016; accepted 18 April 2016
[11] IvetaNováková, KarelMikulica, "Properties of concrete with partial replacement of natural aggregate by recycled concrete aggregates from precast production", International Conference on Ecology and new Building materials and products, ICEBMP 2016.